

Object Oriented Design Quality Metrics

References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [jqassistant](#)
- [notebook walks through examples for integrating various packages with Neo4j](#)
- [OO Design Quality Metrics](#)
- [A Validation of Martin's Metric](#)
- [Neo4j Python Driver](#)

Incoming Dependencies

Incoming dependencies are also denoted as "Fan-in", "Afferent Coupling" or "in-degree". These are the ones that use the listed package.

If these packages get changed, the incoming dependencies might be affected by the change. The more incoming dependencies, the harder it gets to change the code without the need to adapt the dependent code ("rigid code"). Even worse, it might affect the behavior of the dependent code in an unwanted way ("fragile code").

Since Java Packages are organized hierarchically, incoming dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 1a

- Show the top 20 Java Packages with the most incoming dependencies
- Set the "incomingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incomingDependenciesIncludingSubpackages
0	axon-messaging-4.12.2	org.axonframework.messaging	messaging	10051	29462	
1	axon-messaging-4.12.2	org.axonframework.eventhandling	eventhandling	6120	26302	
2	axon-messaging-4.12.2	org.axonframework.commandhandling	commandhandling	1945	6341	
3	axon-messaging-4.12.2	org.axonframework.serialization	serialization	1771	7230	
4	axon-messaging-4.12.2	org.axonframework.messaging.annotation	annotation	1111	4088	
5	axon-messaging-4.12.2	org.axonframework.common	common	1066	2541	
6	axon-messaging-4.12.2	org.axonframework.tracing	tracing	635	2923	
7	axon-messaging-4.12.2	org.axonframework.queryhandling	queryhandling	436	1748	
8	axon-configuration-4.12.2	org.axonframework.config	config	428	3220	
9	axon-messaging-4.12.2	org.axonframework.common.transaction	transaction	368	1103	
10	axon-modelling-4.12.2	org.axonframework.modelling.command	command	276	947	
11	axon-modelling-4.12.2	org.axonframework.modelling.saga	saga	263	1142	
12	axon-messaging-4.12.2	org.axonframework.messaging.unitofwork	unitofwork	260	1102	
13	axon-messaging-4.12.2	org.axonframework.deadline	deadline	256	1252	
14	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.jdbc	jdbc	235	1288	
15	axon-messaging-4.12.2	org.axonframework.messaging.deadletter	deadletter	206	1164	
16	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore	eventstore	205	816	
17	axon-messaging-4.12.2	org.axonframework.monitoring	monitoring	199	531	
18	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing	eventsourcing	148	580	
19	axon-server-connector-4.12.2	org.axonframework.axonserver.connector	connector	129	635	

Table 1b

- Show the top 20 Java Packages including their sub-packages with the most incoming dependencies
- Set the property "incomingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incomingDe
0	axon-messaging-4.12.2	org.axonframework.messaging	messaging	8918	28144	
1	axon-messaging-4.12.2	org.axonframework.eventhandling	eventhandling	3836	14381	
2	axon-messaging-4.12.2	org.axonframework.serialization	serialization	1550	6282	
3	axon-messaging-4.12.2	org.axonframework.common	common	1482	4139	
4	axon-messaging-4.12.2	org.axonframework.commandhandling	commandhandling	1058	4017	
5	axon-messaging-4.12.2	org.axonframework.messaging.annotation	annotation	994	3745	
6	axon-messaging-4.12.2	org.axonframework.tracing	tracing	565	2627	
7	axon-messaging-4.12.2	org.axonframework.common.transaction	transaction	362	1089	
8	axon-configuration-4.12.2	org.axonframework.config	config	302	1972	
9	axon-messaging-4.12.2	org.axonframework.queryhandling	queryhandling	273	1055	
10	axon-messaging-4.12.2	org.axonframework.messaging.unitofwork	unitofwork	223	885	
11	axon-messaging-4.12.2	org.axonframework.monitoring	monitoring	189	501	
12	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing	eventsourcing	173	645	
13	axon-messaging-4.12.2	org.axonframework.messaging.deadletter	deadletter	173	1030	
14	axon-modelling-4.12.2	org.axonframework.modelling.command	command	162	535	
15	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore	eventstore	144	464	
16	axon-messaging-4.12.2	org.axonframework.deadline	deadline	137	773	
17	axon-messaging-4.12.2	org.axonframework.lifecycle	lifecycle	98	271	
18	axon-modelling-4.12.2	org.axonframework.modelling.saga	saga	97	277	
19	axon-messaging-4.12.2	org.axonframework.messaging.responsetypes	responsetypes	90	372	

Outgoing Dependencies

Outgoing dependencies are also denoted as "Fan-out", "Efferent Coupling" or "out-degree". These are the ones that are used by the listed package.

Code from other packages and libraries you're depending on (outgoing) might change over time. The more outgoing changes, the more likely and frequently code changes are needed. This involves time and effort which can be reduced by automation of tests and version updates. Automated tests are crucial to reveal updates, that change the behavior of the

code unexpectedly (“fragile code”). As soon as more effort is required, keeping up becomes difficult (“rigid code”). Not being able to use a newer version might not only restrict features, it can get problematic if there are security issues. This might force you to take “fast but ugly” solutions into account which further increases technical dept.

Since Java Packages are organized hierarchically, outgoing dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 2a

- Show the top 20 Java Packages with the most outgoing dependencies
- Set the "outgoingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgoingDependenciesCount
0	axon-configuration-4.12.2	org.axonframework.config	config	10010	38497	1
1	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.autoconfig	autoconfig	6475	23556	1
2	axon-test-4.12.2	org.axonframework.test.aggregate	aggregate	2210	7978	1
3	axon-messaging-4.12.2	org.axonframework.eventhandling	eventhandling	1563	6457	1
4	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.query	query	1544	7134	1
5	axon-disruptor-4.12.2	org.axonframework.disruptor.commandhandling	commandhandling	1453	5180	1
6	axon-test-4.12.2	org.axonframework.test.saga	saga	1419	4142	1
7	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.jdbc	jdbc	1340	6712	1
8	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.event.axon	axon	1178	4270	1
9	axon-messaging-4.12.2	org.axonframework.queryhandling	queryhandling	1141	5254	1
10	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing	eventsourcing	1009	3348	1
11	axon-messaging-4.12.2	org.axonframework.eventhandling.pooled	pooled	962	4483	1
12	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.command	command	932	3074	1
13	axon-modelling-4.12.2	org.axonframework.modelling.command	command	847	3239	1
14	axon-modelling-4.12.2	org.axonframework.modelling.command.inspection	inspection	793	3001	1
15	axon-server-connector-4.12.2	org.axonframework.axonserver.connector	connector	763	3983	1
16	axon-messaging-4.12.2	org.axonframework.commandhandling.distributed	distributed	738	2227	1
17	axon-messaging-4.12.2	org.axonframework.commandhandling	commandhandling	691	1832	1
18	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore	eventstore	613	1981	1
19	axon-messaging-4.12.2	org.axonframework.eventhandling.deadletter.jdbc	jdbc	560	3518	1

Table 2b

- Show the top 20 Java Packages including their sub-packages with the most outgoing dependencies
- Set the property "outgoingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgoingDependenciesWeight
0	axon-configuration-4.12.2	org.axonframework.config	config	601	1855	
1	axon-messaging-4.12.2	org.axonframework.eventhandling	eventhandling	588	2124	
2	axon-server-connector-4.12.2	org.axonframework.axonserver.connector	connector	416	1529	
3	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.autoconfig	autoconfig	409	1343	
4	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing	eventsourcing	380	1319	
5	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot	springboot	367	1185	
6	axon-test-4.12.2	org.axonframework.test	test	273	793	
7	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore	eventstore	245	914	
8	axon-modelling-4.12.2	org.axonframework.modelling.command	command	232	754	
9	axon-messaging-4.12.2	org.axonframework.eventhandling.deadletter	deadletter	207	996	
10	axon-messaging-4.12.2	org.axonframework.commandhandling	commandhandling	196	485	
11	axon-messaging-4.12.2	org.axonframework.messaging	messaging	193	499	
12	axon-messaging-4.12.2	org.axonframework.deadline	deadline	190	611	
13	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.query	query	179	707	
14	axon-messaging-4.12.2	org.axonframework.queryhandling	queryhandling	177	614	
15	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.event	event	167	548	
16	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.event.axon	axon	167	555	
17	axon-modelling-4.12.2	org.axonframework.modelling.saga	saga	155	474	
18	axon-test-4.12.2	org.axonframework.test.aggregate	aggregate	149	548	
19	axon-disruptor-4.12.2	org.axonframework.disruptor.commandhandling	commandhandling	130	436	

Instability

$$Instability = \frac{Outgoing\ Dependencies}{Outgoing\ Dependencies + Incoming\ Dependencies}$$

Instability is expressed as the ratio of the number of outgoing dependencies of a module (i.e., the number of packages that depend on it) to the total number of dependencies (i.e., the sum of incoming and outgoing dependencies).

Small values near zero indicate low *Instability*. With no outgoing but some incoming dependencies the *Instability* is zero which is denoted as maximally stable. Such code units are more rigid and difficult to change without impacting other parts of the system. If they are changed less because of that, they are considered stable.

Conversely, high values approaching one indicate high *Instability*. With some outgoing dependencies but no incoming ones the *Instability* is denoted as maximally unstable. Such code units are easier to change without affecting other modules, making them more flexible and less prone to cascading changes throughout the system. If they are changed more often because of that, they are considered unstable.

Since Java Packages are organized hierarchically, *Instability* can be calculated for every package in isolation or by including all of its sub-packages.

Table 3a

- Show the top 20 Java Packages with the lowest *Instability*
- Set the property "instability" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityPackag
0	axon-messaging-4.12.2	org.axonframework.messaging	messaging	0.013060	0.086849	0.180723	0.0869
1	axon-messaging-4.12.2	org.axonframework.common.transaction	transaction	0.016043	0.043011	0.000000	0.0303
2	axon-messaging-4.12.2	org.axonframework.common	common	0.021120	0.038462	0.000000	0.0109
3	axon-messaging-4.12.2	org.axonframework.eventhandling.scheduling	scheduling	0.074074	0.111111	0.000000	0.1818
4	axon-messaging-4.12.2	org.axonframework.lifecycle	lifecycle	0.087719	0.179487	0.000000	0.1200
5	axon-messaging-4.12.2	org.axonframework.serialization	serialization	0.090395	0.201754	0.291667	0.1800
6	axon-messaging-4.12.2	org.axonframework.monitoring	monitoring	0.095455	0.145833	0.333333	0.2142
7	axon-messaging-4.12.2	org.axonframework.common.annotation	annotation	0.120000	0.120000	0.000000	0.1666
8	axon-messaging-4.12.2	org.axonframework.common.stream	stream	0.131579	0.150000	0.000000	0.1111
9	axon-messaging-4.12.2	org.axonframework.tracing	tracing	0.161162	0.198529	0.480000	0.2727
10	axon-messaging-4.12.2	org.axonframework.updates.detection	detection	0.166667	0.250000	0.000000	0.2500
11	axon-messaging-4.12.2	org.axonframework.eventhandling	eventhandling	0.203436	0.324544	0.495652	0.2318
12	axon-messaging-4.12.2	org.axonframework.messaging.annotation	annotation	0.212057	0.292793	0.406250	0.2058
13	axon-messaging-4.12.2	org.axonframework.updates.api	api	0.217391	0.333333	0.000000	0.2500
14	axon-messaging-4.12.2	org.axonframework.common.jpa	jpa	0.240000	0.217391	1.000000	0.2500
15	axon-messaging-4.12.2	org.axonframework.common.legacyjpa	legacyjpa	0.260870	0.238095	1.000000	0.2727
16	axon-messaging-4.12.2	org.axonframework.commandhandling	commandhandling	0.262140	0.345794	0.600000	0.2903
17	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot	springboot	0.294737	0.460000	0.000000	0.4285
18	axon-messaging-4.12.2	org.axonframework.messaging.correlation	correlation	0.304348	0.200000	0.400000	0.2857
19	axon-messaging-4.12.2	org.axonframework.serialization.upcasting	upcasting	0.312500	0.083333	0.000000	0.3333

Table 3b

- Show the top 20 Java Packages including their sub-packages with the lowest *Instability*
- Set the property "instabilityIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityPackage
0	axon-messaging-4.12.2	org.axonframework.common	common	0.003362	0.011933	0.000000	0.020000
1	axon-messaging-4.12.2	org.axonframework.messaging	messaging	0.021183	0.075610	0.402878	0.130000
2	axon-messaging-4.12.2	org.axonframework.serialization	serialization	0.033063	0.154839	0.181818	0.180000
3	axon-messaging-4.12.2	org.axonframework.lifecycle	lifecycle	0.039216	0.066667	0.000000	0.080000
4	axon-messaging-4.12.2	org.axonframework.monitoring	monitoring	0.040609	0.052632	0.454545	0.160000
5	axon-messaging-4.12.2	org.axonframework.tracing	tracing	0.055184	0.141414	0.187500	0.270000
6	axon-messaging-4.12.2	org.axonframework.messaging.annotation	annotation	0.071028	0.151515	0.424242	0.180000
7	axon-messaging-4.12.2	org.axonframework.common.annotation	annotation	0.086957	0.086957	0.000000	0.100000
8	axon-messaging-4.12.2	org.axonframework.common.lock	lock	0.107143	0.142857	0.000000	0.140000
9	axon-messaging-4.12.2	org.axonframework.messaging.unitofwork	unitofwork	0.111554	0.100000	0.375000	0.090000
10	axon-messaging-4.12.2	org.axonframework.messaging.responsetypes	responsetypes	0.126214	0.333333	0.000000	0.500000
11	axon-messaging-4.12.2	org.axonframework.eventhandling	eventhandling	0.132911	0.375887	0.452174	0.370000
12	axon-messaging-4.12.2	org.axonframework.common.jdbc	jdbc	0.150943	0.212121	0.000000	0.260000
13	axon-messaging-4.12.2	org.axonframework.common.property	property	0.153846	0.222222	0.000000	0.200000
14	axon-messaging-4.12.2	org.axonframework.commandhandling	commandhandling	0.156300	0.478571	0.512195	0.430000
15	axon-messaging-4.12.2	org.axonframework.common.jpa	jpa	0.157895	0.117647	0.000000	0.200000
16	axon-messaging-4.12.2	org.axonframework.updates.detection	detection	0.166667	0.250000	0.000000	0.250000
17	axon-messaging-4.12.2	org.axonframework.common.legacyjpa	legacyjpa	0.176471	0.133333	0.000000	0.220000
18	axon-messaging-4.12.2	org.axonframework.messaging.deadletter	deadletter	0.183962	0.272727	0.466667	0.330000
19	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.snapshotting	snapshotting	0.187500	0.250000	0.500000	0.330000

Abstractness

$$Abstractness = \frac{\text{abstract classes in category}}{\text{total number of classes in category}}$$

Package *Abstractness* is expressed as the ratio of the number of abstract classes and interfaces to the total number of classes of a package.

Zero *Abstractness* means that there are no abstract types or interfaces in the package. On the other hand, a value of one means that there are only abstract types.

Since Java Packages are organized hierarchically, *Abstractness* can be calculated for every package in isolation or by including all of its sub-packages.

Table 4a

- Show the top 30 packages with the lowest *Abstractness*
- Set the property "abstractness" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
0	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.autoconfigure	autoconfig	0.0	0	41
1	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot	springboot	0.0	0	28
2	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.legacyjpa	legacyjpa	0.0	0	10
3	axon-messaging-4.12.2	org.axonframework.messaging.timeout	timeout	0.0	0	8
4	axon-messaging-4.12.2	org.axonframework.commandhandling.distributed.commandfilter	commandfilter	0.0	0	7
5	axon-messaging-4.12.2	org.axonframework.deadline.dbscheduler	dbscheduler	0.0	0	7
6	axon-messaging-4.12.2	org.axonframework.eventhandling.scheduling.dbscheduler	dbscheduler	0.0	0	7
7	axon-messaging-4.12.2	org.axonframework.serialization.json	json	0.0	0	7
8	axon-messaging-4.12.2	org.axonframework.serialization.xml	xml	0.0	0	7
9	axon-messaging-4.12.2	org.axonframework.tracing.attributes	attributes	0.0	0	6
10	axon-messaging-4.12.2	org.axonframework.updates.api	api	0.0	0	6
11	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.query.subscription	subscription	0.0	0	6
12	axon-tracing-opentelemetry-4.12.2	org.axonframework.tracing.opentelemetry	opentelemetry	0.0	0	5
13	axon-messaging-4.12.2	org.axonframework.eventhandling.deadletter	deadletter	0.0	0	5
14	axon-messaging-4.12.2	org.axonframework.serialization.converters	converters	0.0	0	5
15	axon-test-4.12.2	org.axonframework.test.server	server	0.0	0	4
16	axon-messaging-4.12.2	org.axonframework.commandhandling.callbacks	callbacks	0.0	0	4
17	axon-messaging-4.12.2	org.axonframework.deadline.quartz	quartz	0.0	0	4
18	axon-messaging-4.12.2	org.axonframework.eventhandling.scheduling.java	java	0.0	0	4
19	axon-messaging-4.12.2	org.axonframework.eventhandling.tokenstore.jpa	jpa	0.0	0	4
20	axon-messaging-4.12.2	org.axonframework.updates.detection	detection	0.0	0	4
21	axon-messaging-4.12.2	org.axonframework.eventhandling.scheduling.jobrunr	jobrunr	0.0	0	3
22	axon-messaging-4.12.2	org.axonframework.util	util	0.0	0	3
23	axon-modelling-4.12.2	org.axonframework.modelling.saga.repository.legacyjpa	legacyjpa	0.0	0	3
24	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.event.util	util	0.0	0	3
25	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.processor	processor	0.0	0	3
26	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.inmemory	inmemory	0.0	0	2
27	axon-messaging-4.12.2	org.axonframework.eventhandling.tokenstore.inmemory	inmemory	0.0	0	2
28	axon-messaging-4.12.2	org.axonframework.eventhandling.tokenstore.legacyjpa	legacyjpa	0.0	0	2
29	axon-messaging-4.12.2	org.axonframework.messaging.interceptors.legacyvalidation	legacyvalidation	0.0	0	2

Table 4b

- Show the top 30 packages with the highest *Abstractness* and number of Java Types

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
144	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.jdbc...	statements	1.000000	15	15
145	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.actuator	actuator	1.000000	1	1
143	axon-messaging-4.12.2	org.axonframework.serialization.upcasting	upcasting	0.833333	5	6
140	axon-messaging-4.12.2	org.axonframework.common.annotation	annotation	0.666667	2	3
141	axon-messaging-4.12.2	org.axonframework.common.stream	stream	0.666667	2	3
142	axon-messaging-4.12.2	org.axonframework.eventhandling.scheduling	scheduling	0.666667	2	3
139	axon-messaging-4.12.2	org.axonframework.common	common	0.642857	18	28
136	axon-messaging-4.12.2	org.axonframework.messaging	messaging	0.600000	21	35
137	axon-messaging-4.12.2	org.axonframework.lifecycle	lifecycle	0.600000	6	10
138	axon-messaging-4.12.2	org.axonframework.eventhandling.gateway	gateway	0.600000	3	5
135	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.util	util	0.555556	5	9
132	axon-messaging-4.12.2	org.axonframework.serialization.upcasting.event	event	0.500000	6	12
133	axon-messaging-4.12.2	org.axonframework.common.transaction	transaction	0.500000	2	4
134	axon-modelling-4.12.2	org.axonframework.modelling.saga.metamodel	metamodel	0.500000	2	4
130	axon-messaging-4.12.2	org.axonframework.common.jdbc	jdbc	0.444444	8	18
131	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.conflictresolu...	conflictresolution	0.444444	4	9
129	axon-configuration-4.12.2	org.axonframework.config	config	0.441860	19	43
128	axon-modelling-4.12.2	org.axonframework.modelling.command	command	0.410714	23	56
127	axon-messaging-4.12.2	org.axonframework.eventhandling	eventhandling	0.410000	41	100
122	axon-test-4.12.2	org.axonframework.test.utils	utils	0.400000	2	5
123	axon-test-4.12.2	org.axonframework.test.eventscheduler	eventscheduler	0.400000	2	5
124	axon-messaging-4.12.2	org.axonframework.deadline.annotation	annotation	0.400000	2	5
125	axon-messaging-4.12.2	org.axonframework.queryhandling.registration	registration	0.400000	2	5
126	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.heartbeat	heartbeat	0.400000	2	5
121	axon-modelling-4.12.2	org.axonframework.modelling.saga	saga	0.393939	13	33
118	axon-messaging-4.12.2	org.axonframework.commandhandling	commandhandling	0.375000	12	32
119	axon-messaging-4.12.2	org.axonframework.messaging.interceptors	interceptors	0.375000	3	8
120	axon-messaging-4.12.2	org.axonframework.messaging.responsetypes	responsetypes	0.375000	3	8
117	axon-messaging-4.12.2	org.axonframework.messaging.deadletter	deadletter	0.368421	7	19
116	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore	eventstore	0.354839	11	31

Table 4c

- Show the top 30 packages including their sub-packages with the highest package depth and lowest *Abstractness*

- Set the property "abstractnessIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSubpa
0	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.autoconfig	autoconfig	0.0	0	43	
1	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.heartbe...	connection	0.0	0	2	
2	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.leg...	legacyjpa	0.0	0	10	
3	axon-messaging-4.12.2	org.axonframework.messaging.timeout	timeout	0.0	0	8	
4	axon-messaging-4.12.2	org.axonframework.commandhandling.distributed....	commandfilter	0.0	0	7	
5	axon-messaging-4.12.2	org.axonframework.deadline.dbscheduler	dbscheduler	0.0	0	7	
6	axon-messaging-4.12.2	org.axonframework.eventhandling.scheduling.dbs...	dbscheduler	0.0	0	7	
7	axon-messaging-4.12.2	org.axonframework.serialization.json	json	0.0	0	7	
8	axon-messaging-4.12.2	org.axonframework.serialization.xml	xml	0.0	0	7	
9	axon-messaging-4.12.2	org.axonframework.tracing.attributes	attributes	0.0	0	6	
10	axon-messaging-4.12.2	org.axonframework.updates.api	api	0.0	0	6	
11	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.query.s...	subscription	0.0	0	6	
12	axon-tracing-opentelemetry-4.12.2	org.axonframework.tracing.opentelemetry	opentelemetry	0.0	0	5	
13	axon-messaging-4.12.2	org.axonframework.serialization.converters	converters	0.0	0	5	
14	axon-test-4.12.2	org.axonframework.test.server	server	0.0	0	4	
15	axon-messaging-4.12.2	org.axonframework.commandhandling.callbacks	callbacks	0.0	0	4	
16	axon-messaging-4.12.2	org.axonframework.deadline.quartz	quartz	0.0	0	4	
17	axon-messaging-4.12.2	org.axonframework.eventhandling.scheduling.java	java	0.0	0	4	
18	axon-messaging-4.12.2	org.axonframework.eventhandling.tokenstore.jpa	jpa	0.0	0	4	
19	axon-messaging-4.12.2	org.axonframework.updates.detection	detection	0.0	0	4	
20	axon-messaging-4.12.2	org.axonframework.eventhandling.scheduling.job...	jobrunr	0.0	0	3	
21	axon-messaging-4.12.2	org.axonframework.util	util	0.0	0	3	
22	axon-modelling-4.12.2	org.axonframework.modelling.saga.repository.le...	legacyjpa	0.0	0	3	
23	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.event.util	util	0.0	0	3	
24	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.processor	processor	0.0	0	3	
25	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.inm...	inmemory	0.0	0	2	
26	axon-messaging-	org.axonframework.eventhandling.tokenstore.inm...	inmemory	0.0	0	2	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSubpa
	4.12.2						
27	axon-messaging-4.12.2	org.axonframework.eventhandling.tokenstore.leg...	legacyjpa	0.0	0	2	
28	axon-messaging-4.12.2	org.axonframework.messaging.interceptors.legac...	legacyvalidation	0.0	0	2	
29	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.actuator.axonserver	axonserver	0.0	0	2	

Table 4d

- Show the top 30 packages including their sub-packages with the highest package depth and highest *Abstractness*

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSubpa
123	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.jdbc...	statements	1.000000	15	15	
121	axon-messaging-4.12.2	org.axonframework.common.annotation	annotation	0.666667	2	3	
122	axon-messaging-4.12.2	org.axonframework.common.stream	stream	0.666667	2	3	
120	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.jdbc	jdbc	0.620690	18	29	
119	axon-messaging-4.12.2	org.axonframework.serialization.upcasting	upcasting	0.611111	11	18	
117	axon-messaging-4.12.2	org.axonframework.lifecycle	lifecycle	0.600000	6	10	
118	axon-messaging-4.12.2	org.axonframework.eventhandling.gateway	gateway	0.600000	3	5	
114	axon-messaging-4.12.2	org.axonframework.serialization.upcasting.event	event	0.500000	6	12	
115	axon-messaging-4.12.2	org.axonframework.common.transaction	transaction	0.500000	2	4	
116	axon-modelling-4.12.2	org.axonframework.modelling.saga.metamodel	metamodel	0.500000	2	4	
113	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.util	util	0.454545	5	11	
111	axon-messaging-4.12.2	org.axonframework.common.jdbc	jdbc	0.444444	8	18	
112	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.conflictresolu...	conflictresolution	0.444444	4	9	
110	axon-configuration-4.12.2	org.axonframework.config	config	0.441860	19	43	
109	axon-messaging-4.12.2	org.axonframework.common	common	0.425743	43	101	
105	axon-test-4.12.2	org.axonframework.test.utils	utils	0.400000	2	5	
106	axon-test-4.12.2	org.axonframework.test.eventscheduler	eventscheduler	0.400000	2	5	
107	axon-messaging-4.12.2	org.axonframework.deadline.annotation	annotation	0.400000	2	5	
108	axon-messaging-4.12.2	org.axonframework.queryhandling.registration	registration	0.400000	2	5	
104	axon-modelling-4.12.2	org.axonframework.modelling.command	command	0.380952	32	84	
103	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore	eventstore	0.379747	30	79	
102	axon-messaging-4.12.2	org.axonframework.messaging.responsetypes	responsetypes	0.375000	3	8	
101	axon-messaging-4.12.2	org.axonframework.messaging.deadletter	deadletter	0.368421	7	19	
100	axon-messaging-4.12.2	org.axonframework.messaging	messaging	0.361842	55	152	
99	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing	eventsourcing	0.353383	47	133	
98	axon-modelling-4.12.2	org.axonframework.modelling.command.inspection	inspection	0.346154	9	26	
93	axon-spring-boot-	org.axonframework.actuator	actuator	0.333333	1	3	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSubpa
	autoconfigure-4.12.2						
94	axon-test-4.12.2	org.axonframework.test.saga	saga	0.333333	7	21	
95	axon-messaging-4.12.2	org.axonframework.common.property	property	0.333333	3	9	
96	axon-messaging-4.12.2	org.axonframework.monitoring	monitoring	0.333333	2	6	

Distance from the main sequence

The *main sequence* is a imaginary line that represents a good compromise between *Abstractness* and *Instability*. A high distance to this line may indicate problems. For example is very *stable* (rigid) code with low abstractness hard to change.

Read more details on that in [OO Design Quality Metrics](#) and [Calculate metrics](#).

Table 5a

- Show the top 30 packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-messaging-4.12.2	org.axonframework.serialization.converters	converters	NaN	0.0	NaN	5
1	axon-messaging-4.12.2	org.axonframework.common.io	io	NaN	0.0	NaN	1
2	axon-messaging-4.12.2	org.axonframework.eventhandling.interceptors	interceptors	NaN	0.0	NaN	1
3	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.actuator	actuator	NaN	1.0	NaN	1
4	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.heartbe...	source	NaN	0.0	NaN	1
5	axon-test-4.12.2	org	org	NaN	0.0	NaN	0
6	axon-test-4.12.2	org.axonframework	axonframework	NaN	0.0	NaN	0
7	axon-disruptor-4.12.2	org	org	NaN	0.0	NaN	0
8	axon-disruptor-4.12.2	org.axonframework	axonframework	NaN	0.0	NaN	0
9	axon-disruptor-4.12.2	org.axonframework.disruptor	disruptor	NaN	0.0	NaN	0
10	axon-eventsourcing-4.12.2	org	org	NaN	0.0	NaN	0
11	axon-eventsourcing-4.12.2	org.axonframework	axonframework	NaN	0.0	NaN	0
12	axon-tracing-opentelemetry-4.12.2	org	org	NaN	0.0	NaN	0
13	axon-tracing-opentelemetry-4.12.2	org.axonframework	axonframework	NaN	0.0	NaN	0
14	axon-tracing-opentelemetry-4.12.2	org.axonframework.tracing	tracing	NaN	0.0	NaN	0
15	axon-configuration-4.12.2	org	org	NaN	0.0	NaN	0
16	axon-configuration-4.12.2	org.axonframework	axonframework	NaN	0.0	NaN	0
17	axon-messaging-4.12.2	org	org	NaN	0.0	NaN	0
18	axon-messaging-4.12.2	org.axonframework	axonframework	NaN	0.0	NaN	0
19	axon-spring-boot-autoconfigure-4.12.2	org	org	NaN	0.0	NaN	0
20	axon-spring-boot-autoconfigure-4.12.2	org.axonframework	axonframework	NaN	0.0	NaN	0
21	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.service	service	NaN	0.0	NaN	0
22	axon-modelling-4.12.2	org	org	NaN	0.0	NaN	0
23	axon-modelling-4.12.2	org.axonframework	axonframework	NaN	0.0	NaN	0
24	axon-modelling-4.12.2	org.axonframework.modelling	modelling	NaN	0.0	NaN	0
25	axon-server-connector-4.12.2	org	org	NaN	0.0	NaN	0
26	axon-server-connector-4.12.2	org.axonframework	axonframework	NaN	0.0	NaN	0
27	axon-server-connector-4.12.2	org.axonframework.axonserver	axonserver	NaN	0.0	NaN	0
28	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.event	event	NaN	0.0	NaN	0
29	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.heartbe...	connection	NaN	0.0	NaN	0

Table 5b

- Show the top 30 packages including their sub-packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-messaging-4.12.2	org.axonframework.updates.detection	detection	0.833333	0.000000	0.166667	4
1	axon-server-connector-4.12.2	org.axonframework.axonserver.connector.event.util	util	0.800000	0.000000	0.200000	3
2	axon-messaging-4.12.2	org.axonframework.tracing	tracing	0.744816	0.200000	0.055184	25
3	axon-messaging-4.12.2	org.axonframework.common.lock	lock	0.711039	0.181818	0.107143	11
4	axon-messaging-4.12.2	org.axonframework.serialization	serialization	0.686449	0.280488	0.033063	82
5	axon-test-4.12.2	org.axonframework.test.server	server	0.666667	0.000000	0.333333	4
6	axon-messaging-4.12.2	org.axonframework.common.digest	digest	0.666667	0.000000	0.333333	1
7	axon-messaging-4.12.2	org.axonframework.messaging.annotation	annotation	0.632676	0.296296	0.071028	54
8	axon-messaging-4.12.2	org.axonframework.monitoring	monitoring	0.626058	0.333333	0.040609	6
9	axon-messaging-4.12.2	org.axonframework.messaging	messaging	0.616975	0.361842	0.021183	152
10	axon-test-4.12.2	org.axonframework.test.matchers	matchers	0.606707	0.125000	0.268293	24
11	axon-messaging-4.12.2	org.axonframework.messaging.unitofwork	unitofwork	0.602732	0.285714	0.111554	14
12	axon-messaging-4.12.2	org.axonframework.eventhandling	eventhandling	0.592827	0.274262	0.132911	237
13	axon-messaging-4.12.2	org.axonframework.common.jpa	jpa	0.592105	0.250000	0.157895	4
14	axon-messaging-4.12.2	org.axonframework.common.legacyjpa	legacyjpa	0.573529	0.250000	0.176471	4
15	axon-messaging-4.12.2	org.axonframework.common	common	0.570895	0.425743	0.003362	101
16	axon-messaging-4.12.2	org.axonframework.commandhandling	commandhandling	0.562147	0.281553	0.156300	103
17	axon-messaging-4.12.2	org.axonframework.updates	updates	0.541353	0.142857	0.315789	21
18	axon-messaging-4.12.2	org.axonframework.messaging.correlation	correlation	0.533784	0.250000	0.216216	4
19	axon-messaging-4.12.2	org.axonframework.common.property	property	0.512821	0.333333	0.153846	9
20	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.updates	updates	0.500000	0.000000	0.500000	1
21	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.util.legacyjpa	legacyjpa	0.500000	0.000000	0.500000	1
22	axon-spring-boot-autoconfigure-4.12.2	org.axonframework.springboot.util.jpa	jpa	0.500000	0.000000	0.500000	1
23	axon-messaging-4.12.2	org.axonframework.messaging.responsetypes	responsetypes	0.498786	0.375000	0.126214	8
24	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.eventstore.jdbc...	statements	0.480769	1.000000	0.480769	15
25	axon-eventsourcing-4.12.2	org.axonframework.eventsourcing.snapshotting	snapshotting	0.479167	0.333333	0.187500	3
26	axon-messaging-4.12.2	org.axonframework.serialization.xml	xml	0.466667	0.000000	0.533333	7
27	axon-messaging-4.12.2	org.axonframework.util	util	0.454545	0.000000	0.545455	3
28	axon-messaging-4.12.2	org.axonframework.messaging.deadletter	deadletter	0.447617	0.368421	0.183962	19
29	axon-messaging-4.12.2	org.axonframework.updates.configuration	configuration	0.428571	0.285714	0.285714	7

Abstractness vs. Instability Plot with "Main Sequence" line as reference

- Plot *Abstractness* vs. *Instability* of all packages
- Draw the "main sequence" as dashed green diagonal line
- Scale the packages by the number of types they contain

- Color the packages by their distance to the "main sequence" (blue=near, red=far)

Figure 5a - Packages without their sub-packages

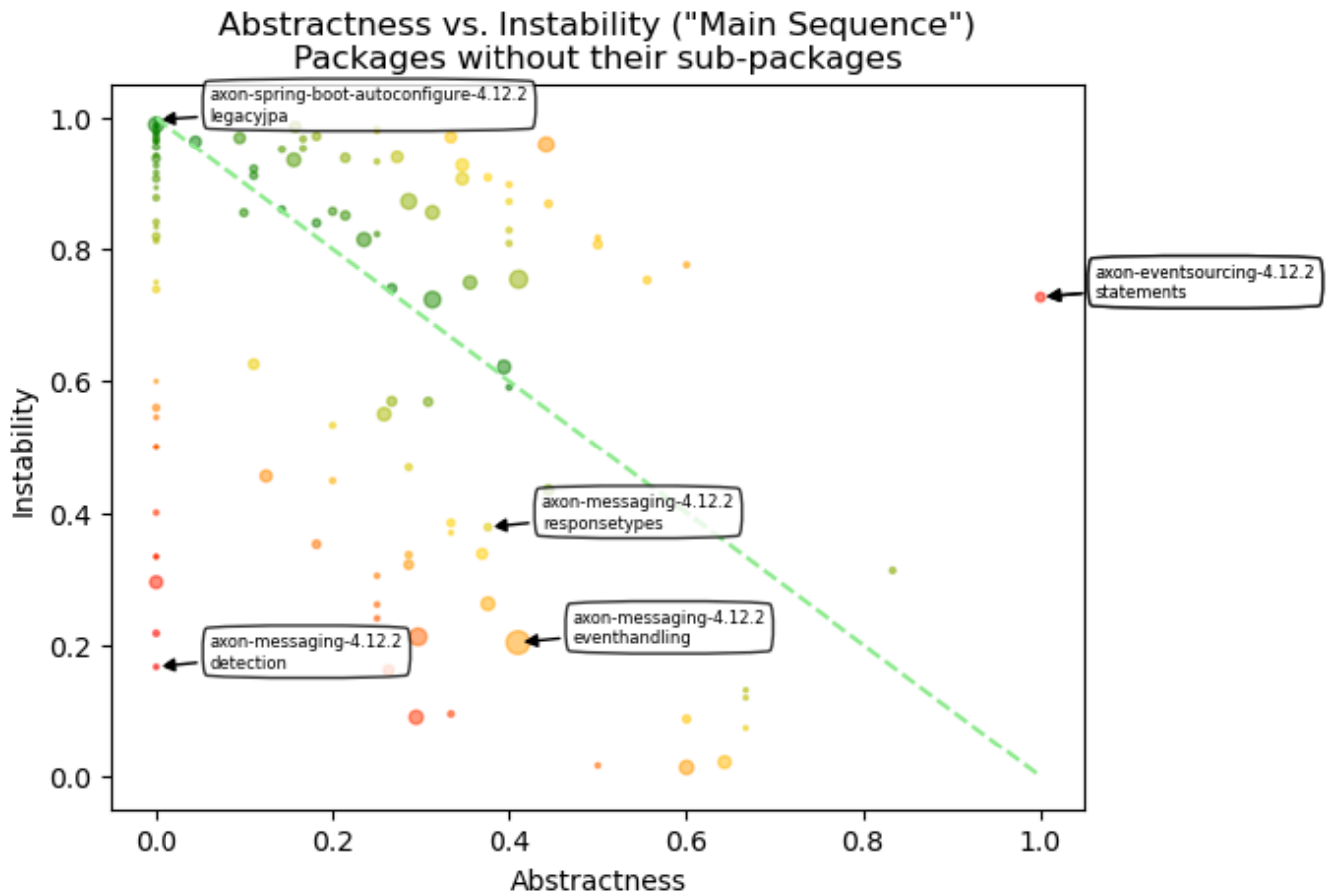


Figure 5b - Packages including their sub-packages

Abstractness vs. Instability ("Main Sequence")
Packages including their sub-packages

